FINAL PROJECT PLAN

Title: Design, Construction and Characterization of a Scanning Acoustic Field Imager / Mapping Instrument using Lock-In Detection

Goal: Building a single-sensor 2D acoustic field mapping instrument capable of measuring both the amplitude and phase of a monochromatic acoustic field using lock-in detection

Minimum success: Clear 2D sound pressure amplitude p(x,y) and separate phase map for one frequency

Expected success: 2D amplitude map R(x,y) and its corresponding phase map phi(x,y) , visualization of the wavefront structure and comparison with simple propagation theory (for a single frequency)

Possible stretch goals: different freqs, different distances, two speakers for interference, quantitative angular directivity analysis, comparing monopole speaker to directional

Equipment:

- <u>Hardware:</u>
  - Speaker (small monopole lab speaker should be fine) to put at center of grid, might need to use a small pedestal to put it at the same height as the microphone
  - Function generator (to drive the speaker at a fixed freq)
  - Lock-in amplifier (for amplitude and phase measurements): keep same sensitivity, time constant and gain while measuring
  - Oscilloscope -> sanity check
  - Microphone (small lab mic preferably)
  - Microphone preamplifier, possibly amplifier for FGEN -> speaker's driving signal
  - Ruler or measuring tape
  - Rigid board or table or anything to define the spacing grid
  - Supports (clamps, books, weights..etc.) to keep positions fixed

- Software:
    - Python for analysis, plots, heat map building (2D color map of amplitude, optional contour lines, can also show radial averages, comparison to theory)

Execution Plan:

Define Grid:
- Choose grid size: 10x10 points
- Choose grid spacing later based on our chosen frequency and thus wavelength
- Choose grid cell size to be the same as the microphone diameter
- Mark coordinates: draw grid on paper/cardboard or tape on the table
- Label axes: x and y, with the center of the grid defined as (0,0)

Pick a frequency:
- Pick a frequency (try sweeping between 2-3 kHz on scope, pick the freq with cleanest sinusoid, largest mic signal, least visible distortion)
- Set FGEN to sine wave and fixed amplitude (don't change throughout experiment)
- Sanity check with oscilloscope: Place mic at (0,0) -> verify sinusoid is there, freq is correct, no saturation of preamp, no clipping or weird noise, reading is stable enough to record (watch V_RMS over 1-2 sec maybe)

CONNECTIONS:
- FGEN main output -> Speaker (through amplifier if needed)
- FGEN sync/ref/2nd output -> SR530 reference input
- Microphone -> Preamp -> SR530 signal input (A input)

Lock-in starting settings:
- Mode: External
- Phase offset: adjust so X is maximized at one reference point, (0,0)

or a corner or edge, so that all other positions are relative
- Sensitivity: 20 mV
- Time const: 100 ms
- Filter slope: 12 dB/oct (default)
- Wait time / points: 0.5 s

Considerations if lock-in adjustments needed:

- **Sensitivity**: X, Y must be well within range, no overload, use lowest sensitivity that doesn't clip, typically 10 - 100 mV range but depends on mic/preamp
- **Time const (tau):** if noisy -> go to 300 ms, don't exceed ~1s unless absolutely necessary, Rule: wait time/point ~ 5(tau)
- **Filter slope:** 18 dB/oct if noise is bad, higher slope = slower response

Test Run:
- Move the mic along a single line (maybe 5 points along x-axis)
- Record X and Y at each point, compute their amplitudes and phases
- Do quick plot with Excel or see if numbers make sense (expected change with position? Repeatable if we measure the same points twice?)
- If works and feels stable, move on to full grid

Full Data Collection:

- **Excel Sheet Data table**:
- coordinates (x,y) for x position and y position (in cm)
- X and Y values from lock-in amplifier
- R (amplitude) for sanity checks (will need to make Excel calculate this)
- Notes (noisy, repeated & averaged...etc.)

- **Measurement:**
- Place mic at (x,y) position to be measured at fixed height (same height as speaker) and note down coordinate values (position)

- Wait a moment for any movement and readings to settle (at least 5tau)
- Record X and Y in table
- If it fluctuates or looks weird, measure 2-3 times and note it (might need to average it, note down any values we averaged)
- Try to keep room as quiet and consistent as possible (maybe no talking or big movements during measurements)

<u>Data analysis:</u>

- Qualitative:
- Where is sound strongest, symmetry, distortions, lobes or nulls?

- Quantitative:
- Compare measured decay with distance to simple model (R(r) proportional to 1/r or 1/r^2 in some region) -> take a line of points along a radial direction, plot R vs r and compare to 1/r trend
- Wavelength lambda = v/f (c~343 m/s)
- Phase vs distance: Plot phi vs r and compare to a straight line with slope k

$$\phi(r) \approx kr + \phi_0 \,, \quad k = \frac{2\pi}{\lambda}$$

(phase should increase linearly with distance (modulo 2π), at least in a simple region away from strong reflections)

<u>Error analysis and limitations:</u>
- Positioning error: how accurately the mic was placed
- Amplitude fluctuations: are readings stable, did we average
- Reflections: walls, table, objects that could've caused interference and distortions
- Background noise: people talking, other equipment, other groups talking...etc.

Can mention: repeating a few points to estimate repeatability and how these uncertainties might distort the map